Ex.no : 9

DATE : 24/09/26

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR.

AIM:

TO implement subnetting in Cisco Packet Tracer Simulator by creating multiple subnets, assigning static IP addresses, configuring routers and establishing communication between different subnets.

PROCEDURE:

1) Open Cisco Packet Tracer and create a new network topology

2) Add 2 routers, 4 switches and 20 PCs and connect them using appropriate cables

3) Consider the network 192.168.1.0/24 and divide it into smaller subnets using the /27 subnet mask.

   255.255.255.224.

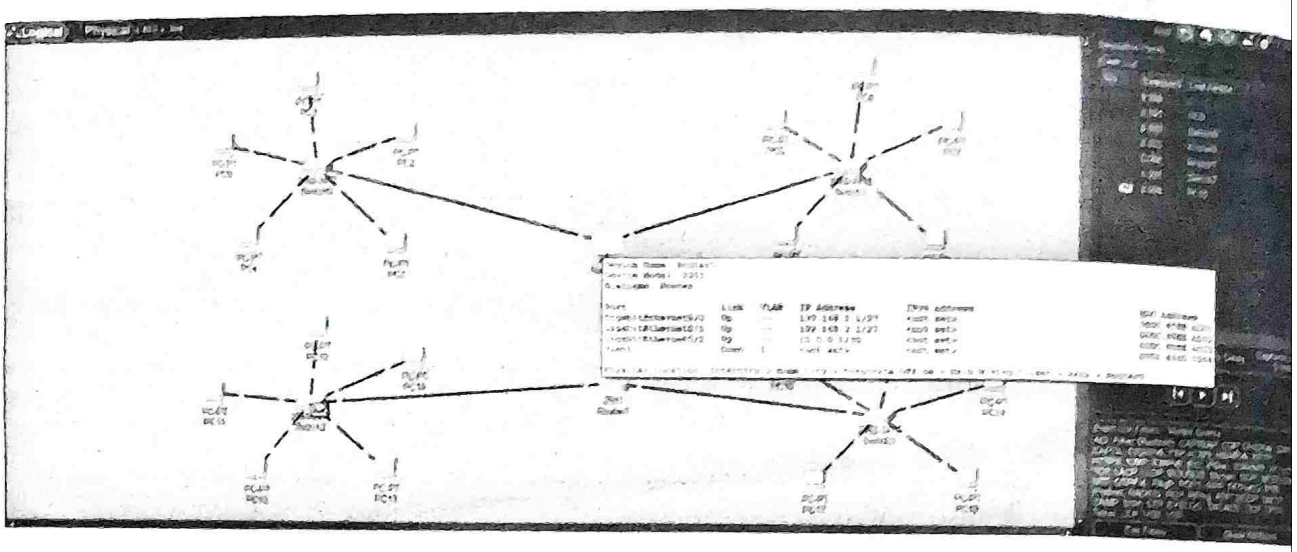
4) Configure Router 0:

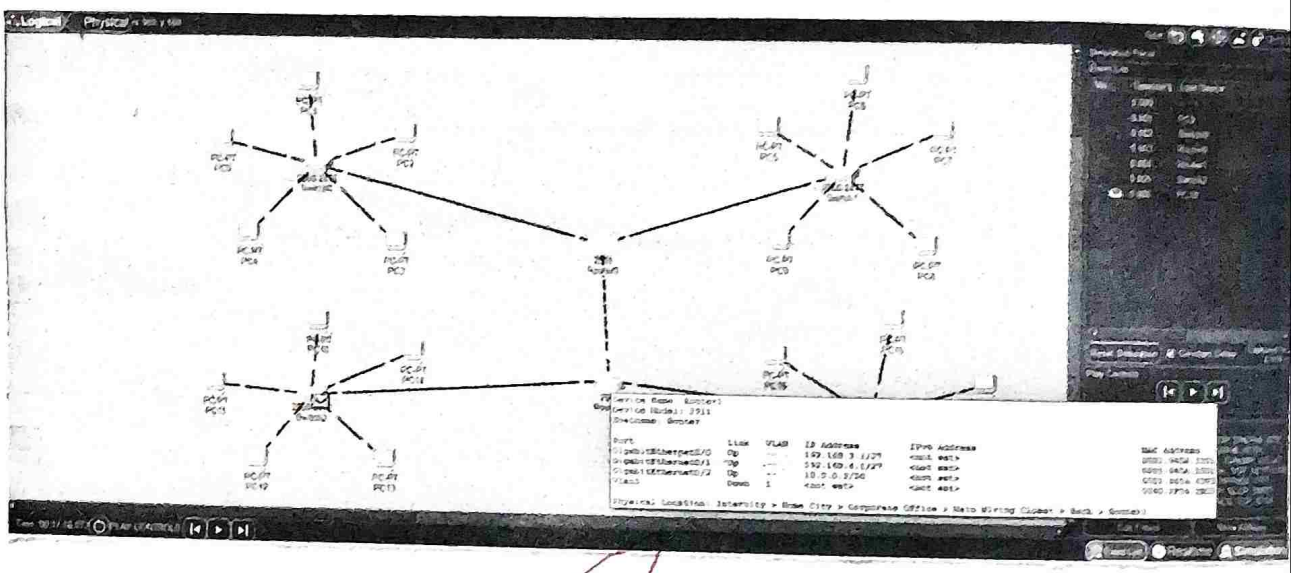
   G0/0 → 192.168.1.1/27
   G0/1 → 192.168.2.1/27
   G0/2 → 10.0.0.1/30

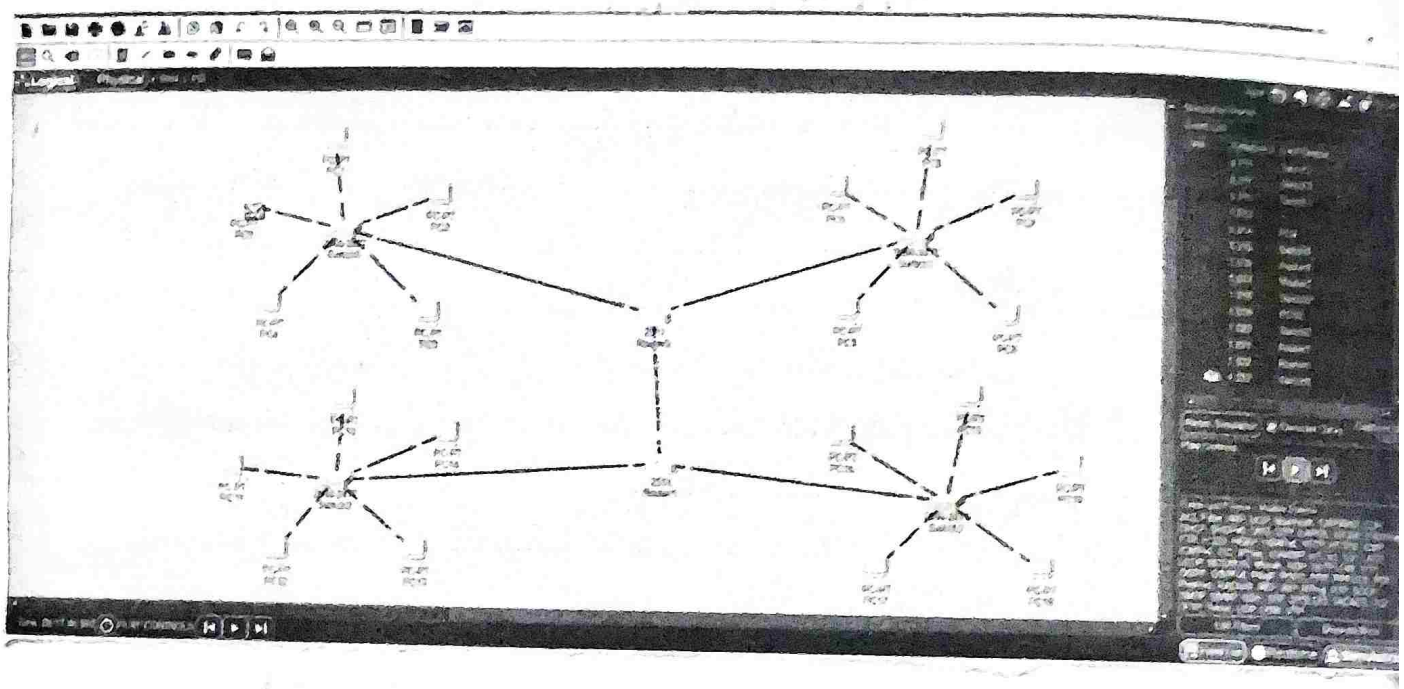
5) Configure Router 1:

   G0/0 → 192.168.3.1/27
   G0/1 → 192.168.4.1/27
   G0/2 → 10.0.0.2/30

6) Enable all router interfaces using the no shutdown command.







7) Assign Static IP addresses to the PCs:

LAN1 : 192.168.1.11 - 192.168.1.15
LAN2 : 192.168.2.11 - 192.168.2.15
LAN3 : 192.168.3.11 - 192.168.3.15
LAN4 : 192.168.4.11 - 192.168.4.15

8) Assign the subnet mask 255.255.255.224 and the appropriate gateway to each PC

9) Configure Static routes between Router 0 and Router 1

Router 0:

Ip route 192.168.3.0 255.255.255.224 10.0.0.2
Ip route 192.168.4.0.255.255.255.224 10.0.0.2

Router 1:

Ip route 192.168.1.0 255.255.255.224 10.0.0.1
Ip route 192.168.2.0 255.255.255.224 10.0.0.1

10) Test communication between PCs in different Subnets using the ping command

11) Switch to Simulation Mode and send a Simple PDU between PCs from different subnets

12) Observe the packet travelling through the switches and routers and verify successful communication.

RESULT:

Thus, the Subnetting was successfully Implemented in Cisco Packet Tracer.